AMRITA VISHWA VIDYAPEETHAM

22MAT230 MATHEMATICS FOR COMPUTING 4

LLM-Guided Regularization of Pseudoinverse for Ill-Posed Signal Reconstruction

TEAM 7 BATCH C

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Problem Statement : Why This Matters?

Real-World Inverse Problems

- True signals cannot be observed directly
- Measurements are distorted and noisy
- Reconstruction requires solving inverse problems

Practical Applications

- Medical imaging (CT, MRI, Ultrasound)
- Image deblurring & restoration
- Signal denoising & communications
- Remote sensing & scientific measurements

Core Challenge

- These problems are ill-posed
- Small noise causes large reconstruction errors
- Direct inversion is unstable

Impact

- Better image quality
- Reliable signal reconstruction
- Efficient algorithms for real systems

Stable and efficient solutions to inverse problems are essential for real-world signal and image recovery.

What Are ill-Posed Inverse Problems?

Forward Problem (Stable)

$$y = Ax$$

- System and signal are known
- Measurements are computed easily
- Small input changes makes small output changes

Inverse Problem (Unstable)

$$x = A^{-1}y^{\delta}$$

- Only measurements are known
- Original signal must be recovered
- Small noise makes large reconstruction error

Why Inverse Problems Are Hard

- Measurements always contain noise
- Matrix A is often ill-conditioned
- Inversion amplifies noise
- Errors grow rapidly in the solution

Mathematical Definition (Hadamard)

A problem is **ill-posed** if it violates any of:

- Solution does not exist
- Solution is not unique
- Solution must depend on the initial conditions.

Understanding Ill-conditioned

$\kappa(A) = \frac{\sigma_{\max}}{\sigma_{\min}}$ (ratio of largest to smallest singular values)

Measures how sensitive the solution is to noise

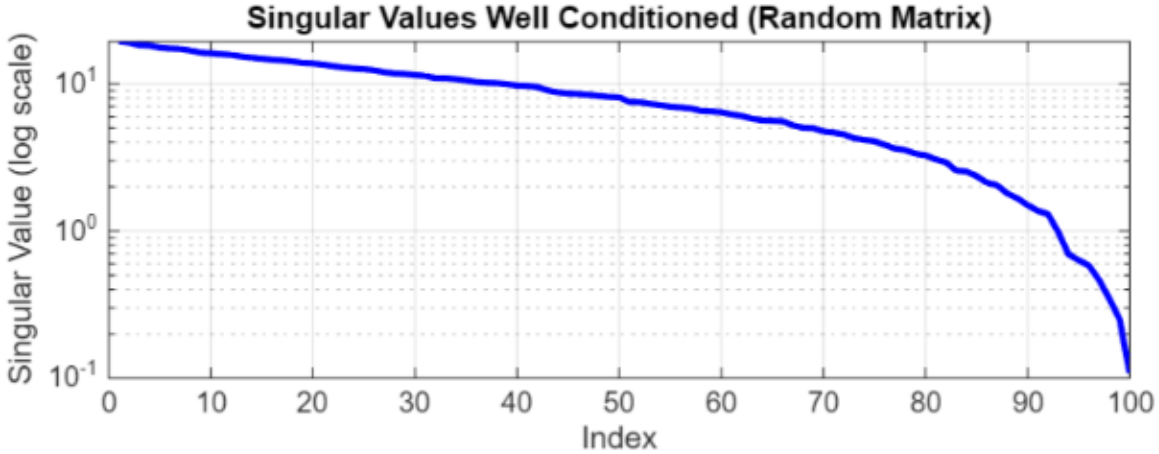
Large $\kappa(A)$ leads to highly unstable inversion

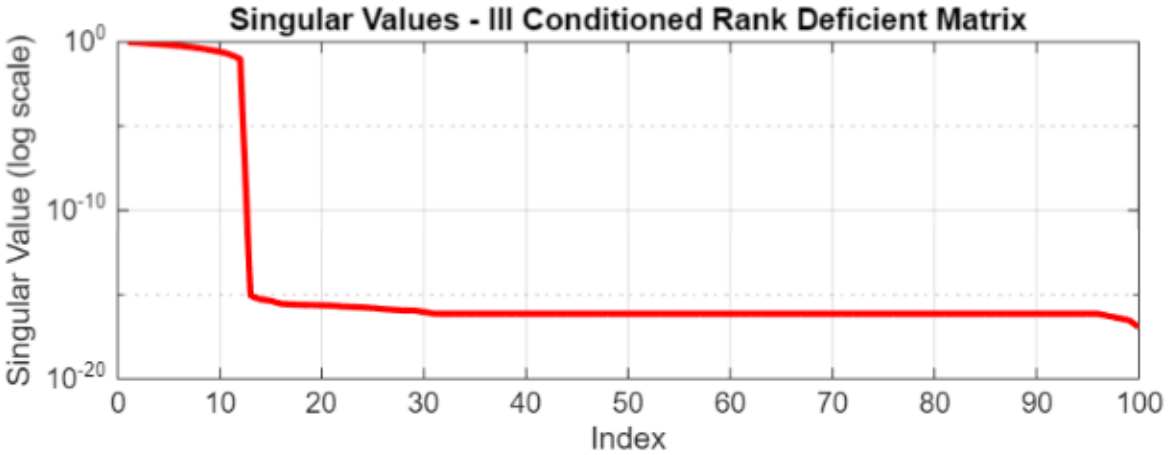
Blur operators have very small

In our project: $\kappa(A) \approx 10^7$

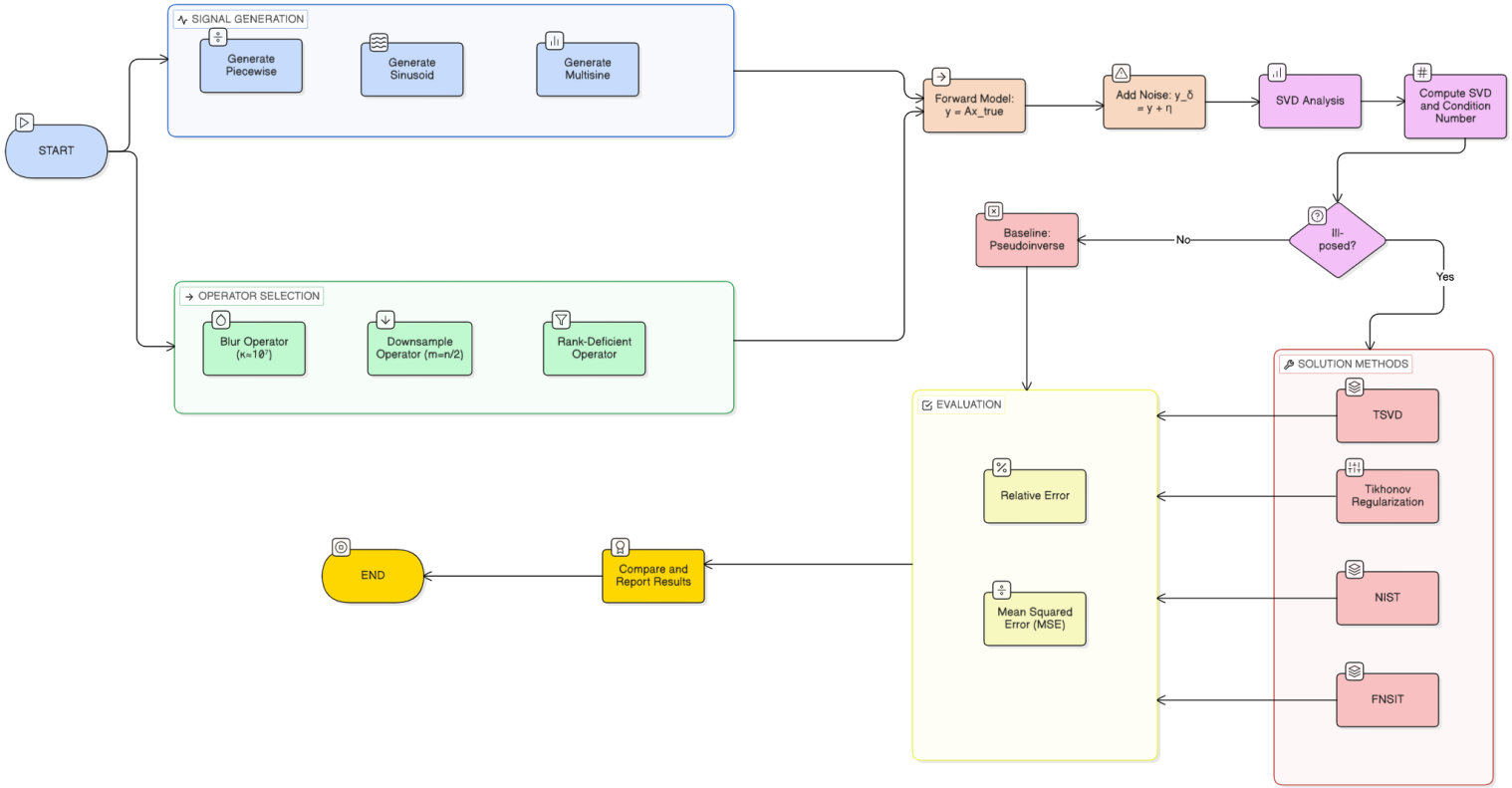
1% noise can become extremely large error in solution

This Explains why direct inversion fails for ill-posed problems


Singular Values Well Conditioned (Random Matrix)
Singular Value (log scale)
10^{1}
10^{0}
10^{-1}
0
10
20
30
40
50
60
70
80
90
100
Index


Singular Values - Ill Conditioned Rank Deficient Matrix
Singular Value (log scale)
10^{0}
10^{-10}
10^{-20}
0
10
20
30
40
50
60
70
80
90
100
Index

METHODOLOGY DIAGRAM



Mathematical Tool for Analysis

- Any matrix $\boldsymbol{A \epsilon R^{(mxn)}}$ can be decomposed as $\boldsymbol{A = U\Sigma V^T}$
- Here $\boldsymbol{U = R^{(mxm)}}$
- Here $\boldsymbol{\Sigma = R^{(mxn)}}$
- Here $\mathrm{V} = \boldsymbol{R}^{(\mathrm{nxn})}$
- $\boldsymbol{\Sigma = \mathrm{diag}(\sigma_1, \sigma_2, \ldots\ldots, \sigma_r, 0, \ldots, 0)}$ where $\boldsymbol{\sigma_1 \geq \sigma_2 \geq \cdots .. \sigma_r > 0}$
- Right Singular Vectors $A^T u_i = \sigma_i v_i$, Columns $v_1, v_2, \ldots\ldots v_n$ forms an orthonormal basis for R^n
- Left Singular Vectors $Av_i = \sigma_i u_i$, Columns $u_1, u_2, \ldots\ldots u_m$ forms an orthonormal basis for R^m
- Condition number: $\kappa(A) = \sigma_1/\sigma_n$ (problem severity measure)

- For Real world problems $y = A\,x + \varepsilon$,Using SVD $y = U\Sigma V^T x + \varepsilon$,
- Rewrite in spectral form we get $U^T y = \Sigma\, V^T x + U^T \varepsilon$,

Method	Uses SVD for
Pseudoinverse	Direct inversion through $\Sigma^{(-1)}$
Tikhonov	Spectral Filtering $\sigma_i \rightarrow \sigma_i/(\sigma_i^2 + \lambda)$
TSVD	Hard Truncation, keeps k largest σ_i
NSIT	Iterative decay of regularization parameter
FNSIT	Fast iterative regularization , it avoids matrix inversion, uses polynomial spectral filter

Pseudo inverse using SVD

- Let $\tilde{y} = U^T y$ and $\tilde{x} = V^T x$
- Then $\tilde{y}_i = \sigma_i \tilde{x}_i + \tilde{\varepsilon}_i$

- Large σ_i, Strong components: Signal recoverable $\tilde{x}_i \approx \tilde{y}_i / \tilde{\varepsilon}_i$ (stable) , Noise effect : $\tilde{\varepsilon}_i$ / $\tilde{y}_i$
- Small σ_i , Weak components: Signal unrecoverable as Noise dominates ,
- Noise effect : $\tilde{\varepsilon}_i / \tilde{y}_i \rightarrow \infty$ as $\sigma_i \rightarrow 0$

- Pseudoinverse using SVD, $A^+ = V\Sigma^+ U^T$
- We get, $x_{\text{pinv}} = A^+ y = V\Sigma^+ U^T y$, $\Sigma^+_{ii} = 1/\sigma_i$
- $x_{pinv} = V\Sigma^+ U^T (Ax_{true} + \epsilon)$ = $x_{true} + V\Sigma^+ U^T \varepsilon$
- $x_{pinv} = \tilde{y}/\sigma_i$, if σ_i $is\ large$
- Error = $V\Sigma^+ U^T \varepsilon$ = $\Sigma(\tilde{\varepsilon}_i/\sigma_i)\ v_i$

Mathematical Setup

Discrete Ill-Posed Inverse Problem

$$y^{\delta} = Ax_{\text{true}} + \eta$$

$A \in R^{m \times n}$: Forward operator (system matrix)

$x_{\text{true}} \in R^{n}$: True signal

$y^{\delta} \in R^{m}$: Noisy measurements

$\eta \sim \mathcal{N}(0, \sigma^2 I)$: Additive Gaussian noise

Our **goal** is to estimate $\hat{x} \approx x_{\text{true}}$ from noisy data

Challenge

- Matrix A is ill-conditioned
- Noise dominates small singular values
- Direct inversion $A^{-1}y^{\delta}$ is unstable and unreliable

Forward Operators – Mathematical Details

1. Gaussian Blur Operator

Models smoothing effects in real systems
Causes loss of high-frequency information
Leads to ill-conditioning

Kernel: $k(i) = \left(\frac{1}{z}\right) exp\left(-\frac{i^2}{2\sigma^2}\right)$

2. Downsampling Operator

Reduces number of measurements
Creates an underdetermined system
Common in signal compression

3. Rank-Deficient Operator

Artificially limits information content
Used to study severely ill-posed problems

$$A = U \Sigma V^T, \qquad \sigma_i = 0 \ for \ i > r$$

Different operators create different types of ill-posedness

Pseudoinverse – The Baseline Failure

Why Naive Inversion Fails in Ill-Posed Problems

For the noisy inverse problem:

$$y^\delta = Ax + \eta$$

Where

$$U = [u_1, \dots, u_m] \quad \text{(left singular vectors)}$$
$$V = [v_1, \dots, v_n] \quad \text{(right singular vectors)}$$
$$\Sigma = diag(\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n)$$

Moore Penrose Pesudo-Inverse

$$A^+ = V\Sigma^{-1}U^T$$

Reconstruction: (Substituting into $\hat{x} = A^+ y^\delta$)

$$\hat{x} = \sum_{i=1}^{n} \frac{u_i^T y^\delta}{\sigma_i} v_i$$

Amplification Factor

$$\text{Amplification factor} = \frac{1}{\sigma_i}$$

Since

$$y^\delta = y + \eta$$

We get:

$$\hat{x} = \sum_{i=1}^{n} \frac{u_i^T y}{\sigma_i} v_i + \sum_{i=1}^{n} \frac{u_i^T \eta}{\sigma_i} v_i$$

The second term represents amplified noise .
In ill-posed problems :

$$\sigma_i \to 0 \quad \Rightarrow \quad \frac{1}{\sigma_i} \to \infty$$

Even tiny noise components dominate the solution.
The reconstruction error satisfies:

$$\frac{|\Delta x|}{|x|} \leq \kappa(A) \frac{|\Delta y|}{|y|}$$

Where

$$\kappa(A) = \frac{\sigma_{max}}{\sigma_{min}} \gg 1$$

For our blur operator:

$$\kappa(A) \approx 10^7$$

Direct pseudoinverse inversion is unstable for noisy ill-posed problems

Regularization Philosophy

In ill posed problems, we face conflict between Accuracy and Stability
Fitting data perfectly $(\lambda \to 0)$: Reconstructs Noise as Signal and the solution diverges (unstable)
Ignoring data $(\lambda \to \infty)$: Smooth Stable solution but misses true signal features
Balancing $(\lambda \simeq \lambda^*)$: Suppress noise amplification and Recovers true signal structure

Ill-posed Problem $y_\delta = Ax + \varepsilon$ (unstable)
Well-posed Problem $\min_x \left|\left|Ax - y_\delta\right|\right|^2 + \lambda R(x)$ (stable)
The regularized problem is **inherently different** from the original, but designed to recover the true signal when noise level and λ are balanced correctly.

Consistency	Solution approaches x_{true} as noise $\to 0$
Stability	Small noise changes → small solution changes
Convergence	With optimal $\lambda(\delta)$, error → 0 as noise → 0

Regularization Philosophy

Problem: Recover image/signal x from blurred noisy image/signal y
Forward: y = Blur(x) + ε, where Blur = A

Unregularized:
$x_{\text{pinv}} = A^{+}y \rightarrow$ Noisy, oscillatory (FAILS)

Regularized:
$$x_\lambda = \underset{x}{\operatorname{argmin}} \left|\left|Ax - y_\delta\right|\right|^2 + \lambda \left|\left|x\right|\right|^2 ; \quad x_\lambda = (A^T A + \lambda I)^{-1} A^T y$$

STEP 1: Choose regularization method
STEP 2: Select regularization parameter λ
STEP 3: Solve optimization problem

Method 1 – Tikhonov Regularization

Smooth Regularization

- Problem with direct inversion - Small singular values amplify noise badly
- Tikhonov Regularization idea - Add a penalty term to stabilize the solution

- Optimization Problem: $\widehat{x_\lambda} = \arg\min_x \{||Ax - y||^2 + \lambda||x||^2\}$

- Closed-Form Solution: $\widehat{x_\lambda} = (A^T A + \lambda I)^{-1} A^T y$

- SVD-Based Form:

$$\widehat{x_\lambda} = \Sigma_{i=1}^{n} \left(f_i^{(Tikh)} * \frac{u_i^T y}{\sigma_i} * v_i \right)$$

- Filter Function: $f_i^{\text{Tikh}} = \dfrac{\sigma_i^2}{\sigma_i^2 + \lambda^2}$

- Key Effect:
  - Large sigma → signal preserved
  - Small → noise suppressed

Tikhonov regularization smoothly damps small singular values to prevent noise explosion.

$$x_\lambda = (A^T A + \lambda I)^{-1} A^T y$$

Singular Value Decomposition:

$$A = U\Sigma V^T$$

Compute A^T
Take transpose:

$$A^T = (U\Sigma V^T)^T$$

Using matrix transpose rule:
So

$$(ABC)^T = C^T B^T A^T$$

$$A^T = V\Sigma^T U^T$$

Since Σ is diagonal:

$$\Sigma^T = \Sigma$$
$$A^T = V\Sigma U^T$$

Compute $A^T A$
Multiplying

$$A^T A = (V\Sigma U^T)(U\Sigma V^T)$$

Group the matrices:

$$A^T A = V\Sigma(U^T U)\Sigma V^T$$

But

$$U^T U = I$$

So

$$A^T A = V\Sigma^2 V^T$$

First part

$$A^T A = V\Sigma^2 V^T$$

So

$$A^T A + \lambda I = V\Sigma^2 V^T + \lambda I$$

Because V is orthogonal:

$$I = VV^T$$

Thus

$$\lambda I = \lambda VV^T$$

Factor V:

$$A^T A + \lambda I = V(\Sigma^2 + \lambda I)V^T$$

Compute the Inverse
Matrix identity: For Orthogonal Matrix,

$$(\mathrm{V}\mathrm{B}V^T)^{-1} = VB^{-1}V^T$$

$$(\mathrm{V}(A^T A + \lambda I)V^T))^{-1} = V(\Sigma^2 + \lambda I))^{-1}V^T$$

$$x_\lambda = V(\Sigma^2 + \lambda I)^{-1}V^T \cdot V\Sigma U^T y$$

Since

$$V^T V = I$$

this simplifies to

$$x_\lambda = V(\Sigma^2 + \lambda I)^{-1}\Sigma U^T y$$

$$x_\lambda = V(\Sigma^2 + \lambda I)^{-1}\Sigma U^T y$$

This is the **spectral form of the Tikhonov solution**.

Tikhonov Filter Analysis $\boldsymbol{f_i(\lambda)} = \sigma_i^2 / \left(\sigma_i^2 + \lambda\right)$

The Tikhonov regularized solution in spectral form: $\widehat{x_\lambda} = f_i(\lambda).\left(\left(u_i^T y_\delta\right)/\sigma_i\right)\mathrm{v}_i$

- Small σ_i : suppressed (σ_i/λ)
- Large σ_i preserved $(1/\sigma_i)$
- Smooth transition, no sharp cutoff

Case 1 :

- Large Singular Values $\sigma_i \gg \lambda$
- $\boldsymbol{f_i(\lambda)} = \sigma_i^2 / \left(\sigma_i^2\right) = 1$ (filter ≈ 1) , Signal components are preserved.
- Amplification: $\sigma_i^2 / \left(\sigma_i^2 + \lambda\right) \cdot (1/\sigma_i) \approx 1/\sigma_i$

Case 2 :

- Small Singular Values $\sigma_i \ll \lambda$
- $\boldsymbol{f_i(\lambda)} \approx \sigma_i^2 / (\lambda) \approx 0$ (filter ≈ 0) , Noise dominated components are suppressed.
- Amplification: $\sigma_i^2 / \left(\sigma_i^2 + \lambda\right) \cdot (1/\sigma_i) \approx \sigma_i/\lambda$

Method 2 : Truncated SVD (TSVD)

•Pseudoinverse fails because **small singular values amplify noise**
•Idea: **Do not invert small singular values**
•TSVD achieves this by **explicit truncation**

Instead of using **all** singular values, we:
Keep only the **largest** $k$singular values
Discard components dominated by noise

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k \gg \sigma_{k+1}, \ldots$$

TSVD Reconstruction Formula
Given SVD:

$$A = U\Sigma V^T \text{ and } y_\delta = \mathrm{Ax}$$

The **TSVD solution** is:

$$\boxed{\hat{x}_k = \sum_{i=1}^{k} \frac{u_i^T y^\delta}{\sigma_i} v_i}$$

- Only First K spectral components are used
- Noise-dominated components are ignored

Filter Function Interpretation
TSVD can be written using a **filter function**:

$$\hat{x}_k = \sum_{i=1}^{n} f_i^{TSVD} \frac{u_i^T y^\delta}{\sigma_i} v_i$$

where:

$$\boxed{f_i^{TSVD} = \begin{cases} 1, & i \leq k \\ 0, & i > k \end{cases}}$$

This is a **hard cutoff** in the spectral domain.
Why TSVD Works
Removes inversion of **small** σ_i
Prevents noise amplification
Retains dominant signal components
Mathematically:

$$\frac{1}{\sigma_i} \textit{ used only when } \sigma_i \textit{ is large}$$

TSVD STABILIZES INVERSION BY DISCARDING NOISE-DOMINATED SINGULAR COMPONENTS

How Do we Find the Value of K?

The truncation parameter $k$controls the balance between **signal recovery** and **noise suppression**.
Small k→ oversmoothed solution (loss of signal details)
Large k→ small singular values amplify noise
Hence, selecting an optimal $k$is essential for stable reconstruction.

To determine the optimal truncation level, we perform a **parameter sweep** over possible values of k:

$$k = 1, 2, 3, \dots, k_{max}$$

For each candidate k, the TSVD reconstruction is computed as

$$\widehat{x_k} = \sum_{i=1}^{k} \frac{u_i^T y_\delta}{\sigma_i} v_i$$

For each k, the reconstruction quality is evaluated using the **relative error**

$$\text{RelErr}(k) = \frac{|x_{\text{true}} - \widehat{x_k}|_2}{|x_{\text{true}}|_2}$$

The optimal truncation parameter is selected as

$$k^* = \arg\min_k \text{RelErr}(k)$$

This value provides the best trade-off between **noise suppression** and **signal preservation**.

Tikhonov vs TSVD Comparison

Filter Function Comparison

- **Key Differences:**
    - Tikhonov**:** Gradual transition, produces smooth solutions
    - TSVD**:** Abrupt transition, preserves strong features
- **When to Use:**
    - Tikhonov**:** General inverse problems, smooth signals
    - TSVD**:** Known dimensionality, feature-rich signals
- **Performance**
    - Both methods achieve ~96% error reduction

Tikhonov smooths the spectrum, TSVD cuts it off.

Method 3 : NSIT – Non-Stationary Iterated Tikhonov

Standard Tikhonov solves

$$x_\lambda = \arg\min_x \| Ax - y_\delta \|^2 + \lambda \| x \|^2$$

Closed form solution:

$$x_\lambda = (A^T A + \lambda I)^{-1} A^T y_\delta$$

Here $\lambda > 0$ is the **regularization parameter** that controls the balance between **data fitting and noise suppression**.

Idea of NIST
Instead of using **one fixed regularization parameter**, NIST **updates the regularization parameter at each iteration**.

$$\lambda_1 > \lambda_2 > \lambda_3 > \cdots > \lambda_k$$

This means:

Early iterations : strong smoothing
Later iterations : recover fine signal details

At iteration k, we solve the regularized problem

$$x_k = \arg\min_x \left(\left|\left|Ax - y_\delta\right|\right|^2 + \lambda_k \left|\left|x - x_{\{k-1\}}\right|\right|^2 \right)$$

$$J(x) = \left|\left|Ax - y_\delta\right|\right|^2 + \lambda_k \left|\left|x - x_{\{k-1\}}\right|\right|^2$$

$$\nabla J(x) = 2A^T(Ax - y_\delta) + 2\lambda_k(x - x_{\{k-1\}})$$

$$A^T A\, x - A^T y_\delta + \lambda_k x - \lambda_k x_{\{k-1\}} = 0$$

$$(A^T A + \lambda_k I)x = A^T y_\delta + \lambda_k x_{\{k-1\}}$$

$$x_k = (A^T A + \lambda_k I)^{-1}(A^T y_\delta + \lambda_k x_{\{k-1\}})$$

Method 3 : NSIT – Non-Stationary Iterated Tikhonov

NIST Iterative Update Rule
The NIST iteration is

$$x_k = \left(A^T A + \lambda_k I\right)^{-1}\left(A^T y_\delta + \lambda_k x_{k-1}\right)$$

$$\lambda_k = \lambda_0\, q^{k-1}, \qquad 0 < q < 1$$

where

- x_k is the **reconstructed signal at iteration** k,
- x_{k-1} is the **previous estimate.**
- $A^T y_\delta$ represents the **information from measured data**.
- $\lambda_k x_{k-1}$ stabilizes the solution using the **previous estimate as prior information**.
- q Decay factor controlling how fast regularization decreases

•Early iterations: **large λ** → strong noise suppression
•Later iterations: **smaller λ** → better signal recovery

Iterative Process

Algorithm:

1.Initialize

$$x_0 = 0$$

2.For $k = 1,2, \ldots$

$$\lambda_k = \lambda_0\, q^{k-1}$$
$$x_k = \left(A^T A + \lambda_k I\right)^{-1}\left(A^T y_\delta + \lambda_k x_{k-1}\right)$$
$$r_k = y_\delta - A x_k$$

3. Stop when residual satisfies **Morozov discrepancy principle**

Stopping Criterion (Morozov Principle)

Iterations stop when the residual satisfies

$$\| \mathrm{r_k} \| \leq \tau\delta$$

Where

- δ is the **noise level in the measurements**
- $\tau > 1$is a **safety factor controlling stopping tolerance**.

Method 4 : FNSIT – Fast Non Stationary Iterated Tikhonov

Objective: The FNIST (Fast Nonstationary iterated Tikhonov) reconstruction method is an iterative algorithm used to solve ill-posed inverse problems.
It improves stability and convergence by combining two correction terms with adaptive regularization.

Algorithm Steps

1. Initialize $x_0 = 0$
2. Compute adaptive parameter α_k
3. Compute residual $r_k = y - Ax_k$
4. Calculate correction terms F_1 and F_2
5. Update solution x_{k+1}
6. Repeat for max iterations

Iteration Parameter

$$\alpha_k = \alpha_0 \cdot q^{k-1}$$

Residual

$$r_k = y - Ax_k$$

First Correction Term

$$F_1 = (A^T A + \alpha_k I)^{-1} A^T r_k$$

Second Correction Term

$$F_2 = \alpha_k (A^T A + \alpha_k I)^{-1} \left(A^T A + \alpha_k^2 I\right)^{-1} A^T r_k$$

Final Update

$$x_{k+1} = x_k + 0.7F_1 + 0.3F_2$$

Error Metrics

Metric 1: Relative Error (Primary Metric)

$$RelErr = \frac{|x_{true} - \hat{x}|_2}{|x_{true}|_2}$$

All metrics use the **Euclidean (ℓ_2) norm**:

$$\| x \|_2 = \sqrt{\sum_{i=1}^{n} x_i^2}$$

Metric 2: Mean Squared Error (MSE)

$$MSE = \frac{1}{n}\sum_{i=1}^{n}\left(x_{true,i} - \hat{x}_i\right)^2$$

Parameter Optimization

A **parameter sweep** is a systematic process of testing multiple values of a parameter to find which value produces the best results.

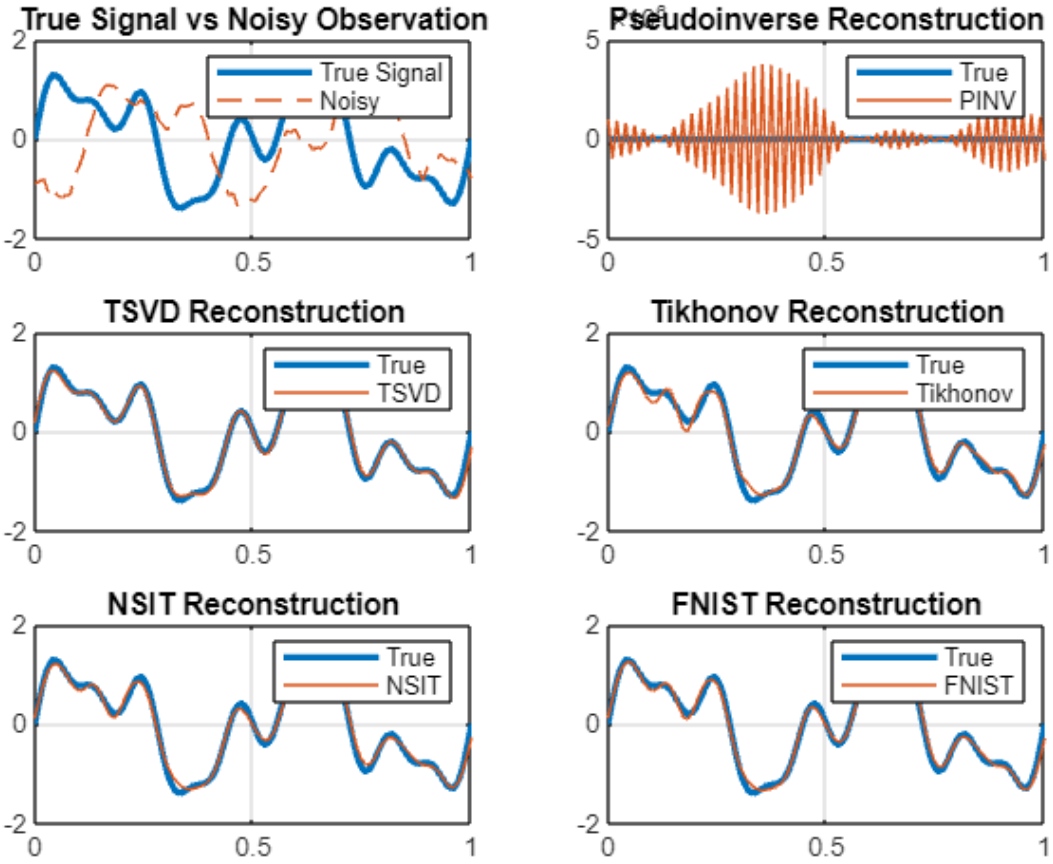
Tikhonov Parameter Sweep Results:
Lambda range: 1.00e-06 to 1.00e+02
Optimal lambda: 5.428675e-02

TSVD Parameter Sweep Results:
k range: 1 to 99
Optimal k: 19

Optimal FNIST Parameters:
alpha0 = 4.9417
q = 0.980
iterations = 28

Results – Signals

```
Pseudoinverse | Rel Error: 914.959750 | MSE: 517987.393743
TSVD          | Rel Error: 0.052696 | MSE: 0.001718
Tikhonov      | Rel Error: 0.132030 | MSE: 0.010786
NSIT          | Rel Error: 0.104887 | MSE: 0.006807
FNIST         | Rel Error: 0.032848 | MSE: 0.000668
```


True Signal vs Noisy Observation
True Signal
Noisy
Pseudoinverse Reconstruction
True
PINV
TSVD Reconstruction
True
TSVD
Tikhonov Reconstruction
True
Tikhonov
NSIT Reconstruction
True
NSIT
FNIST Reconstruction
True
FNIST

Integration With LLM

LLM Integration for Parameter Selection

•Selecting the correct **regularization parameter** is critical for stable signal reconstruction.

•Manual tuning of parameters (λ for Tikhonov, k for TSVD, iterations for NSIT) is **time-consuming and problem-dependent**.

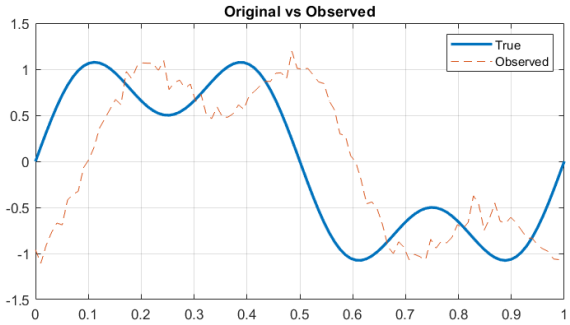
•We integrate a **Large Language Model (LLM)** to assist in **adaptive parameter selection**.

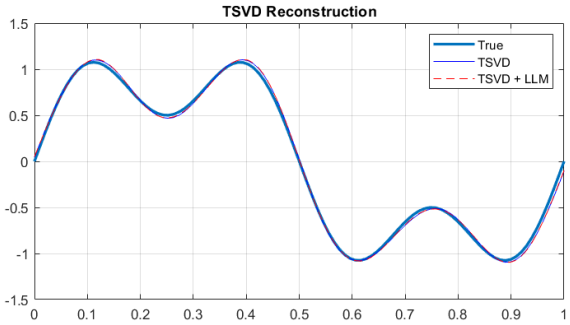
•The LLM analyzes reconstruction information such as **noise level, condition number, and error metrics**.

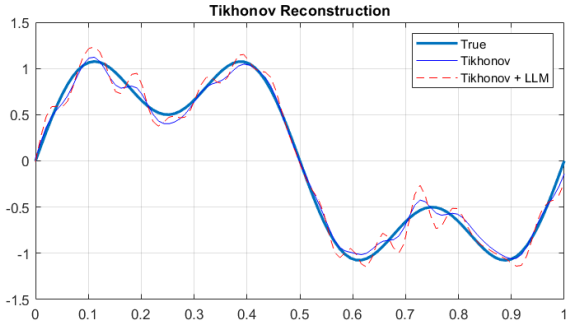
•Based on this information, it suggests **better parameter values** to improve reconstruction quality.

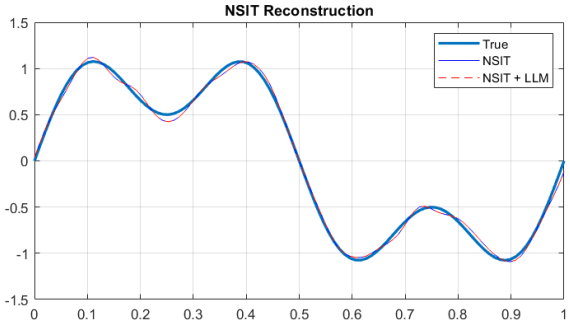
•This reduces manual experimentation and helps achieve **more stable and accurate solutions**.

Key Idea:
The LLM does not solve the inverse problem itself, it **guides the regularization process by selecting better parameters**.









	methods	before_llm	after_llm	improvement
1	"TSVD"	0.0413	0.0412	5.4277e-05
2	"Tikhonov"	0.1426	0.0803	0.0624
3	"NSIT"	0.0503	0.0503	0
4	"FNIST"	2.1574	0.0452	2.1121

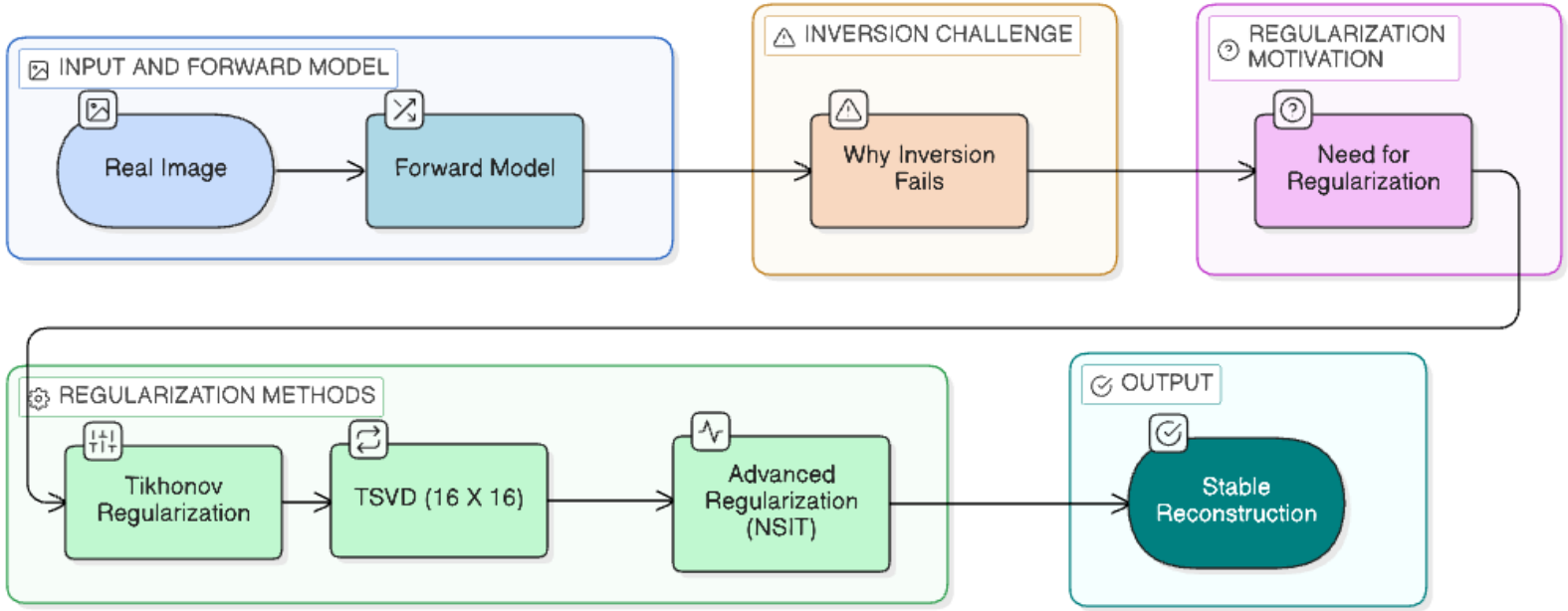
Image Reconstruction

In **Real-world situation ,** We start with a **clean image** (unknown): $x \in \mathbb{R}^n$
Also The imaging system (camera, optics, sensor): blurs the image , introduces noise This gives the **observed image**: $y \in \mathbb{R}^n$

Given y , recover x , This is the problem , $x = A^{-1}y$, Noise ε is small,
but $|| \mathrm{A}^{-1} (y + \varepsilon) - \mathrm{A}^{-1}y || \gg ||\varepsilon||$, Small noise causes huge reconstruction error

The workflow starts from forward modeling, demonstrates the failure of naive inversion, and progressively introduces operator-based regularization methods culminating in NSIT for stable full-image reconstruction.

Image Reconstruction:



Toy Example :

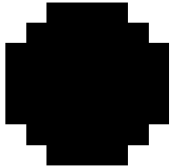
$$y = A\,x + \varepsilon$$

$$x^{\dagger} = A^{\dagger} y = V\Sigma^{-1}U^{T}y = \sum_{i=1}^{r} \frac{u_i^T y}{\sigma_i} v_i$$

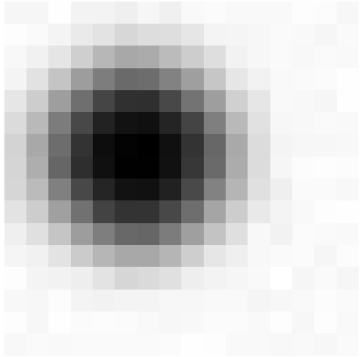
$$\frac{u_i^T \varepsilon}{\sigma_i} \gg 1 \implies \text{reconstruction blows up}$$

$$\text{MSE} = \frac{1}{N}\sum (x_i^{\text{rec}} - x_i^{\text{true}})^2, \quad \text{Relative Error} = \frac{\|x^{\text{rec}} - x^{\text{true}}\|}{\|x^{\text{true}}\|}$$

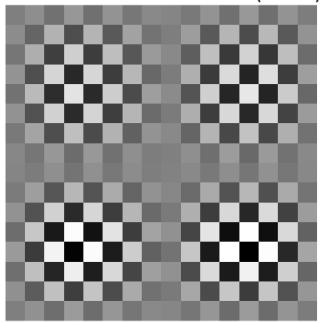
Toy Image (16x16)

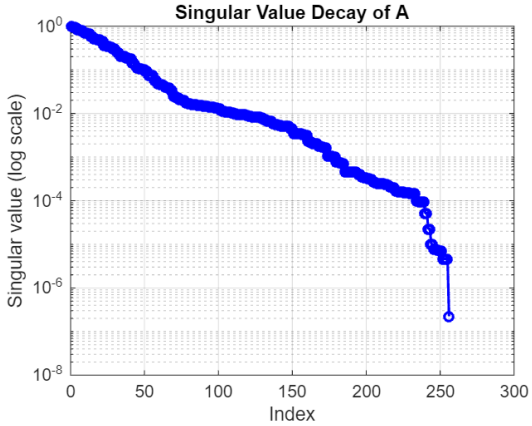


Blurred + Noisy Image y = Ax + ϵ



Pseudoinverse Reconstruction (FAILS!)





Visual Reconstruction Comparison

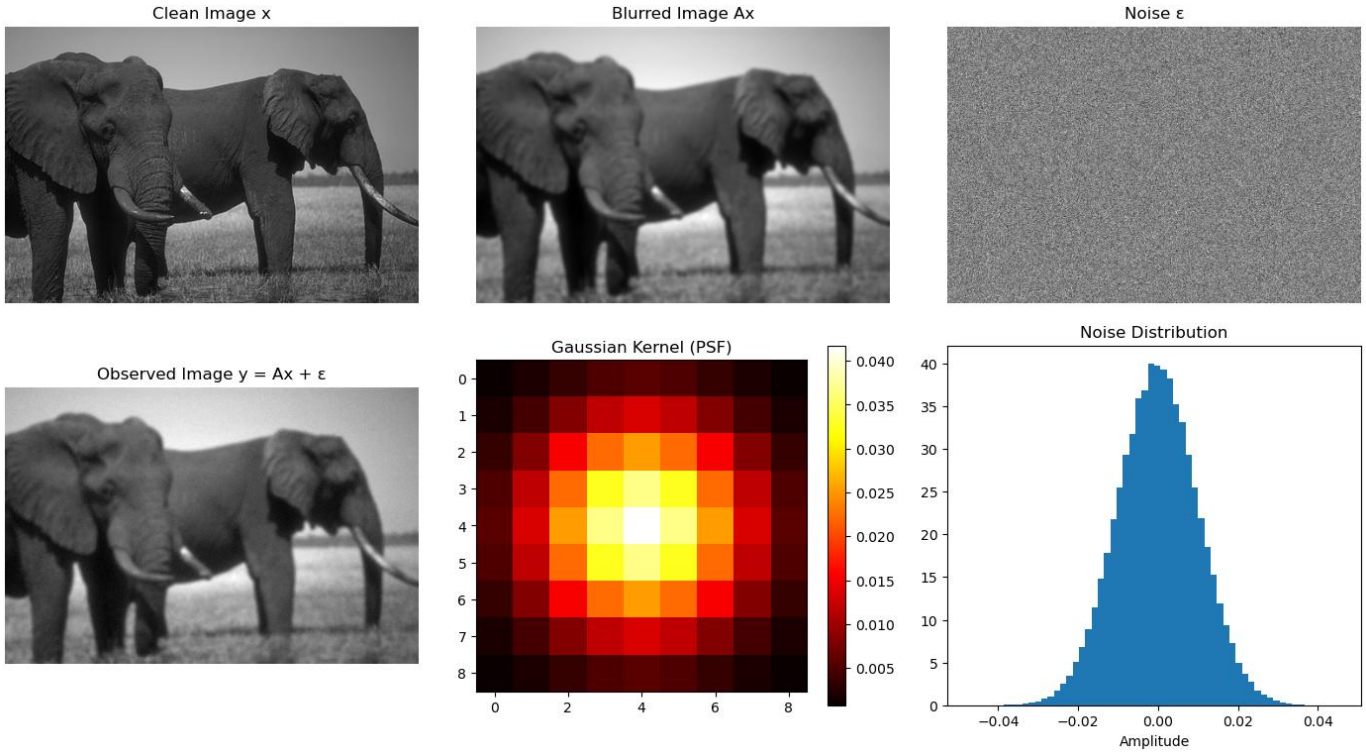


Image Reconstruction Comparison:

Manual vs LLM-Guided Image Reconstruction

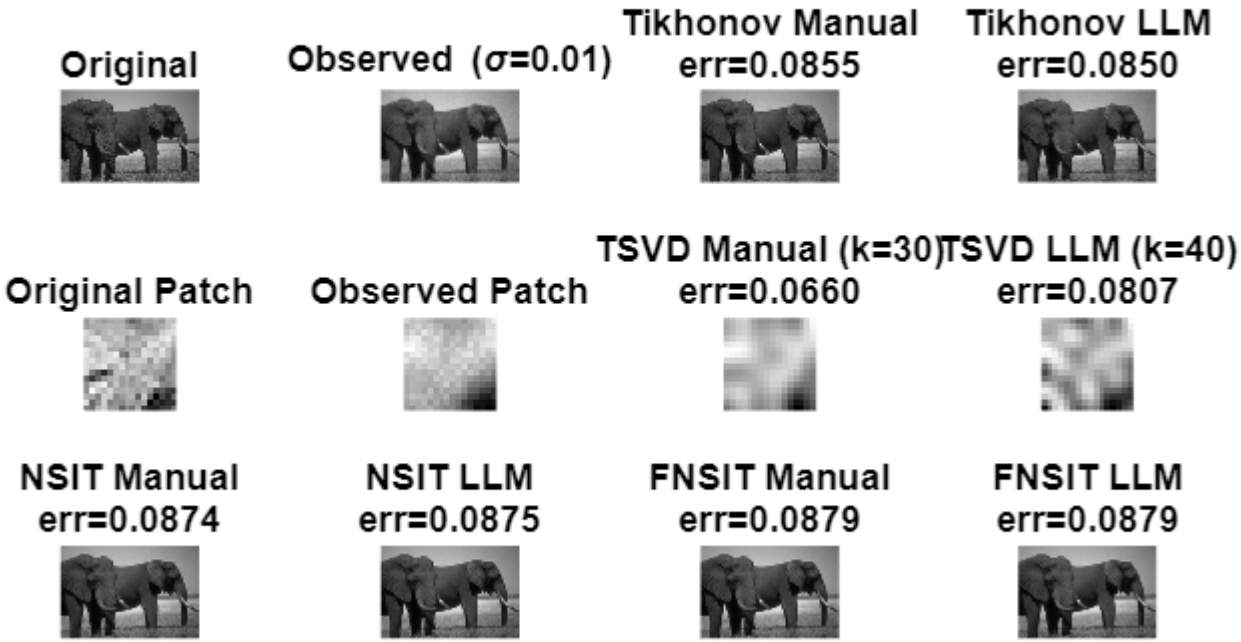


Image Reconstruction Results Obtained:

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============================================================
  COMPARISON: MANUAL vs LLM-GUIDED PARAMETERS
============================================================
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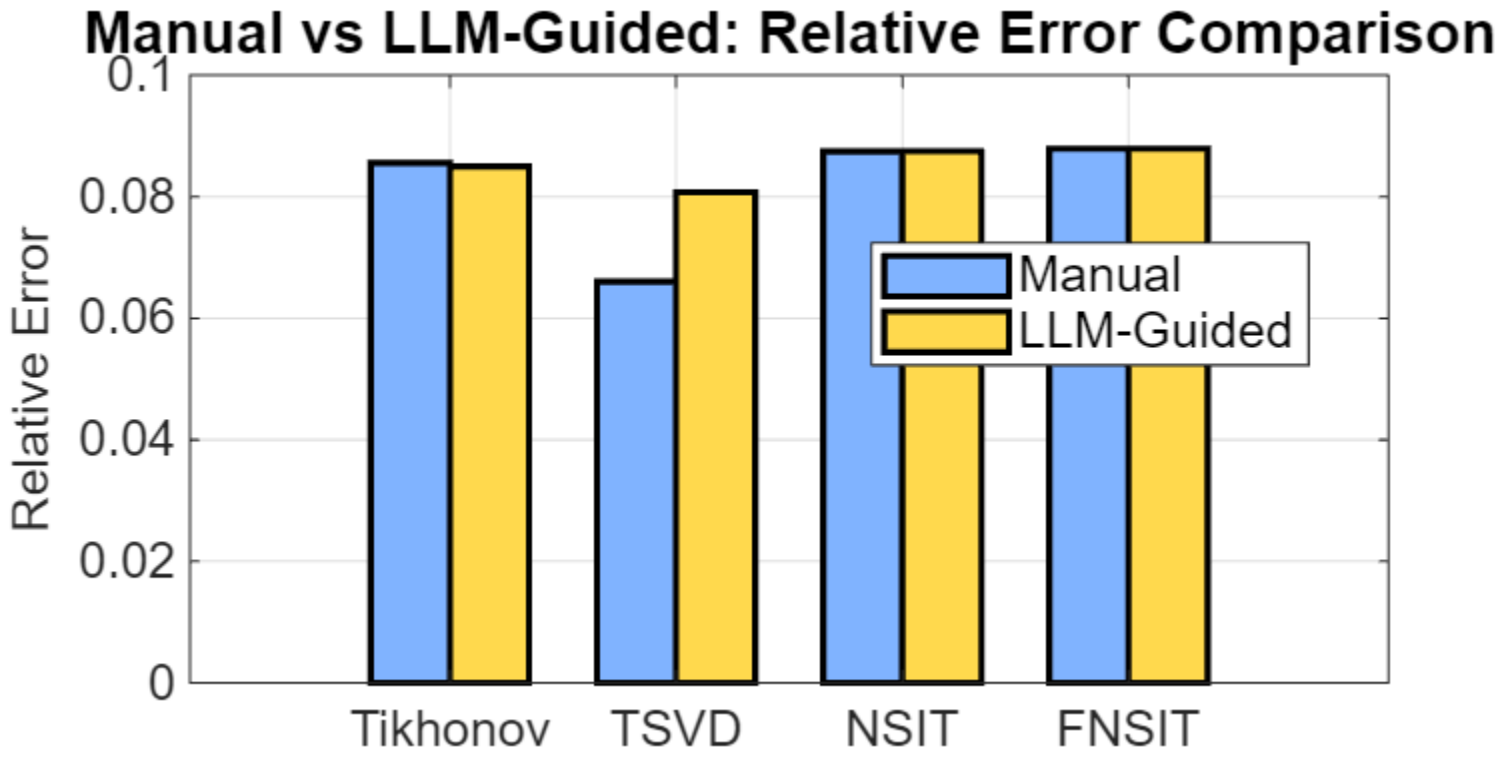
Method	Manual RelErr	LLM RelErr	Manual MSE	LLM MSE
Tikhonov	0.0855	0.0850	0.001315	0.001297
TSVD (patch)	0.0660	0.0807	0.003057	0.004568
NSIT	0.0874	0.0875	0.001373	0.001375
FNSIT	0.0879	0.0879	0.001388	0.001388

Image Reconstruction:



Signal Reconstruction Results Obtained:

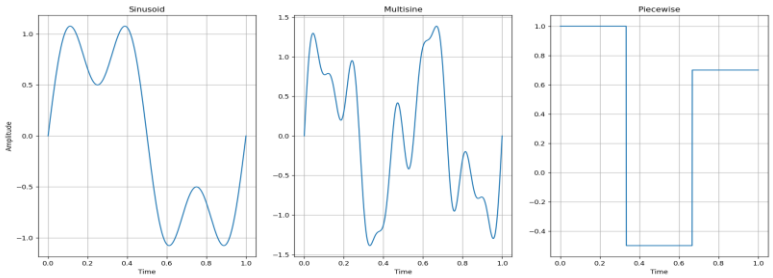
	methods	before_llm	after_llm
1	"TSVD"	0.0369	0.0369
2	"Tikhonov"	0.1134	0.1679
3	"NSIT"	0.0685	0.0680
4	"FNIST"	0.0613	5.4390

Comprehensive Testing Framework

We use **three representative 1D signals**:

- **Sinusoidal signal**
- **Multi-sine signal**
  Superposition of multiple frequencies
- **Piecewise signal**
  Step function with sharp discontinuities

Covers **smooth, oscillatory, and discontinuous** cases



Each signal is distorted using **three operators**:

- **Gaussian Blur Operator**
  Convolution-based smoothing
- **Downsampling Operator**
  Dimension reduction
  Underdetermined system
- **Rank-Deficient Operator**
  Controlled loss of information
  Exact zero singular values

Reconstruction Methods Compared

- **Pseudoinverse (baseline – expected to fail)**
- **Tikhonov regularization**
- **Truncated SVD (TSVD)**
- **NSIT**
- **FNSIT**

Key Takeaways

- Ill-posed inverse problems are highly unstable and cannot be solved by direct inversion
- The pseudoinverse fails due to severe noise amplification from small singular values
- Regularization is mandatory to obtain stable and meaningful reconstructions
- Tikhonov and TSVD effectively suppress noise and reduce reconstruction error by ~96%
- SVD is the core analytical tool, explaining ill-posedness and guiding all regularization methods
- Regularization parameters are critical, balancing noise suppression and signal fidelity
- Iterative methods (NSIT) and FNSIT provide automatic, adaptive regularization without manual tuning
- Proposed methods are robust across signals, operators, and noise levels

References and Resources

[1] Huang, Ce, Li Wang, Minghui Fu, Zhong-Rong Lu, and Yanmao Chen. "A novel iterative integration regularization method for ill-posed inverse problems." Engineering with Computers 37, no. 3 (2021): 1921-1941.

[2] Ji, Kunpu, et al. "An adaptive regularized solution to inverse ill-posed models." IEEE Transactions on Geoscience and Remote Sensing 60 (2022): 1-15.

[3] Alberti, Giovanni S., et al. "Learning the optimal Tikhonov regularizer for inverse problems." Advances in Neural Information Processing Systems 34 (2021): 25205-25216.

[4] Huang, C., Wang, L., Fu, M., Lu, Z.R. and Chen, Y., 2021. A novel iterative integration regularization method for ill-posed inverse problems. Engineering with Computers, 37(3), pp.1921-1941.

Thank You !!